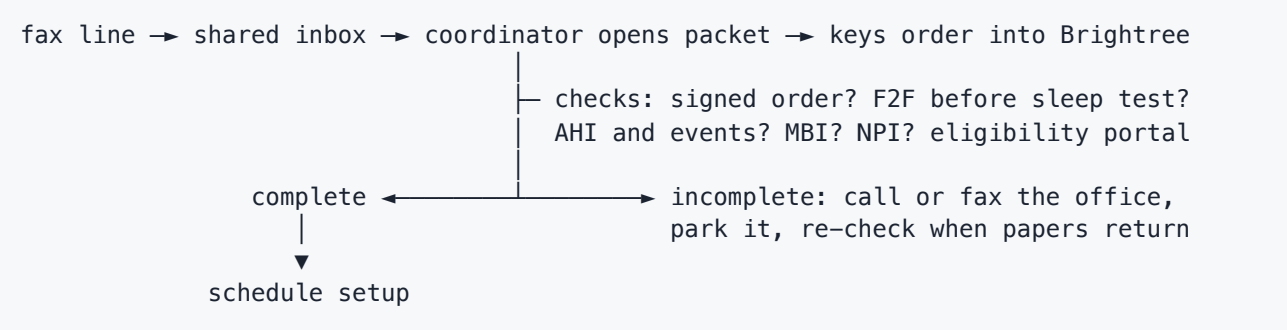
Implementation kit: Cascade Sleep & Respiratory

*Fictional customer. Every number marked **(A)** is an assumption to replace in discovery; every number marked **(M)** is measured on this repo's synthetic eval (`make eval`, 2026-10-26).*

1. Discovery and operational audit

Who they are (A). Regional sleep and respiratory DME supplier, Washington and Oregon. ~1,500 new PAP orders a month, growing into home oxygen. Six intake coordinators. Orders arrive by fax from ~400 referring practices; billing runs in Brightree.

Current intake workflow (A).

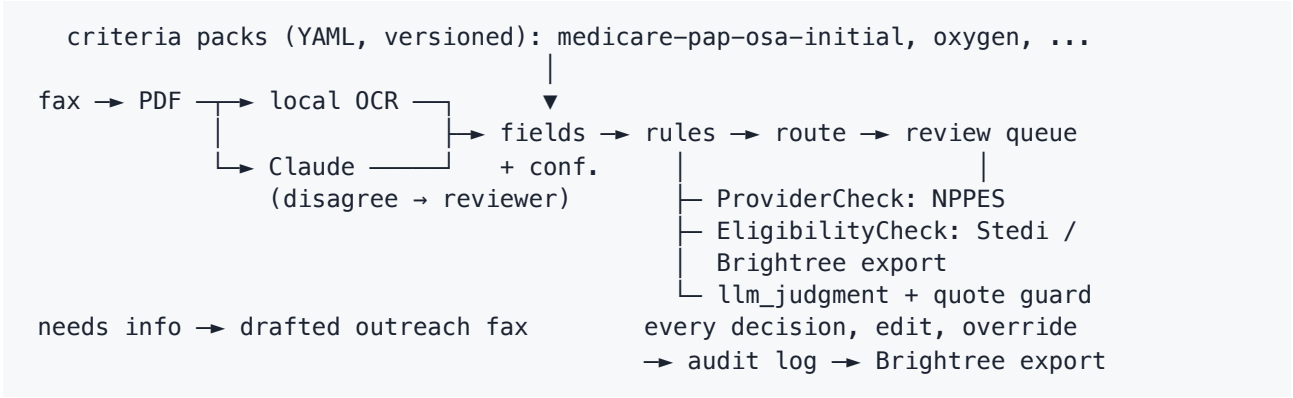


Where the time goes (A). ~18 minutes of hands-on time per order: 4 min finding and ordering pages, 5 min keying, 6 min checking criteria, 3 min eligibility lookup. Incomplete orders add a 10-minute outreach cycle, usually more than once.

Top defect types. The synthetic set plants the defects that drive Medicare PAP documentation errors in published DME MAC review findings: missing or unsigned orders, face-to-face notes that don't predate the sleep test, AHI 5–14 without a documented qualifying condition, missing sleep study pages, and provider or eligibility problems. In discovery, replace the planted mix with 200 of their own recent orders, labeled by a senior coordinator.

Defect (planted)	Share of synthetic orders	Who fixes it
Unsigned order	10%	Referring office (needs info)
F2F after sleep test	10%	Referring office (needs info)
Sleep study page missing	8%	Referring office (needs info)
AHI 5–14, no qualifying condition documented	8%	Reviewer, then office
Invalid / inactive NPI	8%	Reviewer
Eligibility inactive	8%	Reviewer / billing
Key field illegible	5%	Reviewer

2. Solution design



- **Packs:** start with `medicare-pap-osa-initial`; `medicare-oxygen-home` in phase 3. Each rule cites the LCD or documentation article it comes from.
- **Integration points.** Fax line → `POST /orders`. Eligibility: Stedi today; their existing clearinghouse or a Brightree eligibility export plugs in behind `EligibilityCheck` without touching rules. Provider: NPPES behind `ProviderCheck`, with their referring-practice roster checked first. Output: the decision JSON maps onto a Brightree sales order and document attachments.
- **Review workflow.** Queue sorted Human review → Needs info → Ready, oldest first. A coordinator approves or overrides the route; overrides are saved as new eval cases.
- **Thresholds.** Auto-ready only when every field a rule used was read at ≥ the pack's bar (0.92 for PAP, set from the sweep as the lowest bar with zero defective orders auto-routed). Fields below 0.6 make their rule *unverifiable*: a person checks, nothing is guessed.

3. Business case

Formula

```
minutes saved / month = orders × (t_before − t_after)
t_after               = a × t_ready_check + (1 − a) × t_review
FTE hours freed       = minutes saved / 60
FTEs                  = FTE hours freed / 140 productive hours per FTE-month
```

Input	Value	Source
orders / month	1,500	(A)
t_before: hands-on minutes per order today	18	(A)
a: share auto-routed Ready	50%	(M) PAP, at 0.92
t_ready_check: spot check of an auto-ready order	1.5 min	(A)
t_review: reviewer time on a flagged order, with findings and evidence pre-highlighted	6 min	(A), validate in shadow mode
pipeline time per order	about 21 seconds median including OCR, unattended	(M)
model cost per order	$0.068	(M)

Result (with the inputs above): t_after = 3.8 min. Minutes saved per month = 21,375, which is **356 hours, or 2.5 FTEs**, redeployed to outreach and patient setup. Model spend is $101 a month.

Orders cleared faster. Incomplete orders get a drafted, cited request the same hour they arrive instead of after a coordinator reaches them in the inbox.

Denials avoided (A). If 5% of orders currently ship with a documentation defect, every defect the engine catches before dispensing is a denial or recoupment avoided. On the synthetic set it caught 100% of planted defects (M); real-world recall on their own faxes is exactly what phase 0 measures, so size this line with their denial log then.

4. Rollout plan

Phase	What runs	Humans	Exit criteria	Owner
0. Shadow (4 weeks)	Engine routes every order; decisions are logged, not acted on	Coordinators work as today; weekly diff review	≥ 95% defect recall on their labeled orders; zero defective orders the engine would have called Ready	SE + intake lead
1. Auto-route Ready	Orders above the bar skip full review; 10% random spot check	Coordinators review flagged orders in the queue	Spot-check error rate < 1%; reviewer minutes per flagged order ≤ 6	Intake lead
2. Outreach drafts	Needs-info orders get a drafted fax; coordinator sends it	Coordinator edits and sends	≥ 80% of drafts sent without edits; days-to-complete down 30%	Intake lead + practice liaison
3. More packs	Add `medicare-oxygen-home`, then continued-coverage PAP	Same queue	Each pack passes its own synthetic + 100-order shadow eval before go-live	SE

Risks

- *Rule accuracy.* Packs encode real policy; a wrong criterion undercuts trust. Every rule cites its source; a DME billing lead signs off each pack version before go-live.
- *Fax quality worse than synthetic.* Shadow mode on their own faxes sets the real threshold; the confidence bar moves per pack, not in code.
- *Over-trusting automation.* The system never approves or denies; Ready means the documentation is complete. Spot checks never go to zero.
- *PHI.* Production needs BAA-covered model hosting, access control, encryption at rest, and audit retention (see README).